

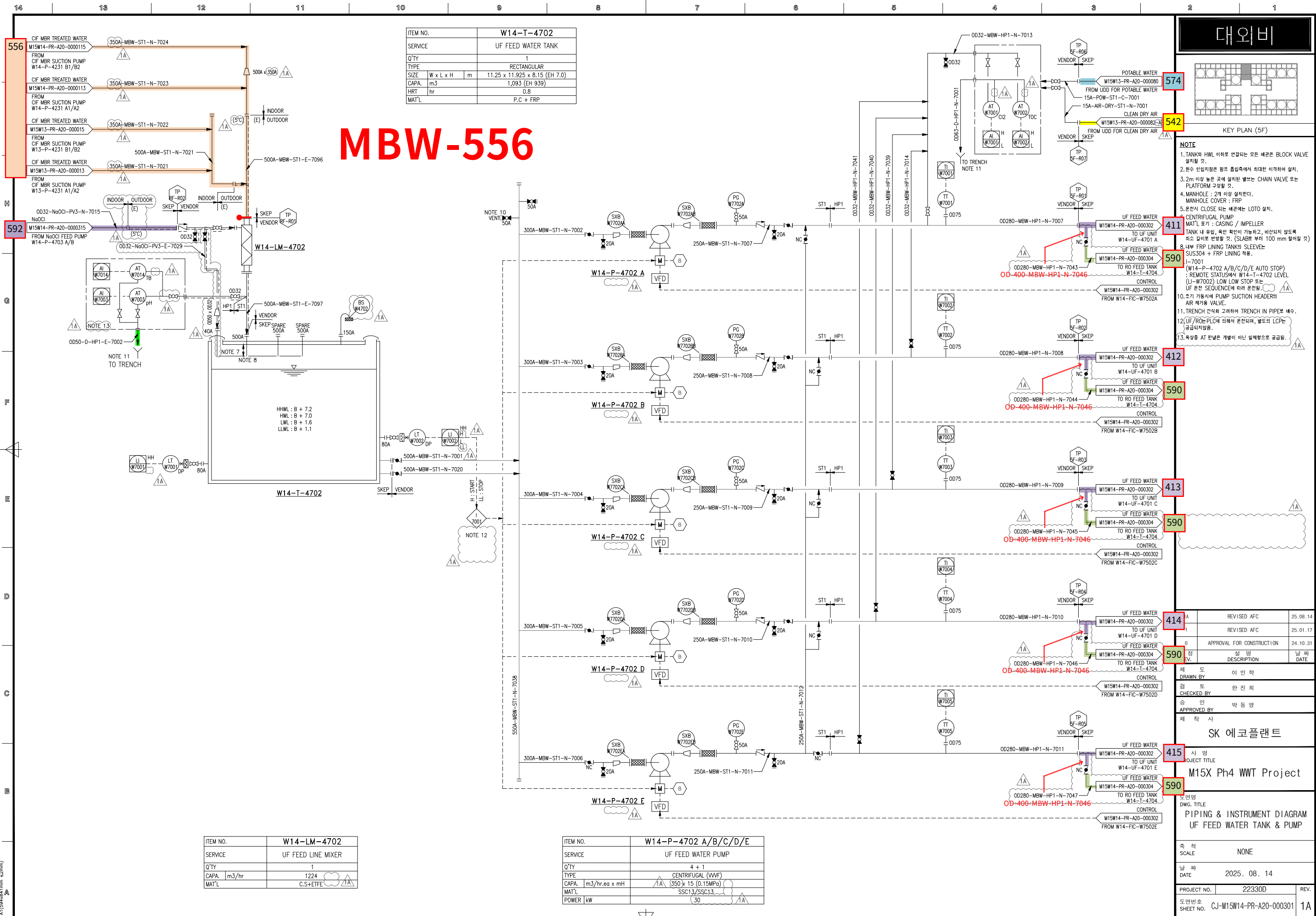
DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

MBW-556

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	556	4FA	400A-MBW-ST1-N-7021-01	1 OF 1	0	CJ-M15W13-PR-A20-000013	1A	
2	556	4FA	400A-MBW-ST1-N-7022-01	1 OF 1	0	CJ-M15W13-PR-A20-000015	1A	
3	556	4FA	400A-MBW-ST1-N-7023-01	1 OF 1	0	CJ-M15W14-PR-A20-000113	1A	
4	556	4FA	400A-MBW-ST1-N-7024-01	1 OF 1	0	CJ-M15W14-PR-A20-000115	1A	
5	556	4FA	500A-MBW-ST1-N-7021-01	1 OF 1	0	CJ-M15W14-PR-A20-000301	1A	
6	556	4FA	500A-MBW-ST1-N-7096-01	1 OF 1	0	CJ-M15W14-PR-A20-000301	1A	
7	556	5FA	500A-MBW-ST1-N-7021-01	1 OF 1	0	CJ-M15W14-PR-A20-000301	1A	
8	556	5FA	500A-MBW-ST1-N-7096-01	1 OF 1	0	CJ-M15W14-PR-A20-000301	1A	
9	556	RFA	500A-MBW-ST1-E-7021-01	1 OF 1	1	CJ-M15W14-PR-A20-000301	1A	
10	556	RFA	500A-MBW-ST1-E-7096-01	1 OF 1	1	CJ-M15W14-PR-A20-000301	1A	



ITEM NO.	W14-AD-4229 B	
SERVICE	CIF MBR 2 nd OXIC TANK AIR DIFFUSER	
Q'TY	(110)	2A
TYPE	MEMBRANE (DISC)	
CAPA. L/min.ea	100	
MAT'L	PP + EPDM	

ITEM NO.	W14-T-4229 B		
SERVICE	CIF MBR 2'nd OXIC TANK		
Q'TY	1		
TYPE	RECTANGULAR		
SIZE	W x L x H m	5.175 x 5.25 x 11.0 (EH 8.8)	
CAPA.	m ³	2A (298.9 (EH 239))	
HRT	min	0.4	
MAT'L	R.C + FRP		

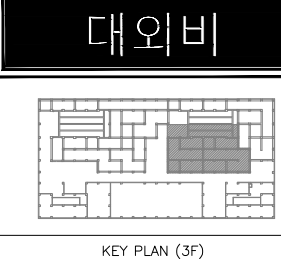
ITEM NO.	W14-T-4230 B		
SERVICE	CIF MBR TANK		
Q'TY	1		
TYPE	RECTANGULAR 		
SIZE	W x L x H	m	2.9 x 22.65 x 6.15 (EH 3.8) 2.85 x 22.65 x 6.15 (EH 3.8)
CAPA.	m ³		801 (EH 495)
HRT	hr		0.8
MAT'L	P.C + FRP		

ITEM NO.		W14-T-4232 B	
SERVICE		CIF MBR CLEANING TANK	
Q'TY		1	
TYPE		RECTANGULAR	
SIZE	W x L x H m	(2.6 x 3.0 x 4.7 (EH 3.2))	
CAPA.	m ³	36.7 (EH 25)	
HRT	min	-	
MAT'L		C.S + FRP	

ITEM NO.	W14-AD-4230 B		
SERVICE	CIF MBR AIR DIFFUSER		
Q'TY	ea	{64}	{2A}
TYPE	MEMBRANE (DISC)		
CAPA.	L/min.ea	{100}	{2A}
MAT'L	PP + EPDM		

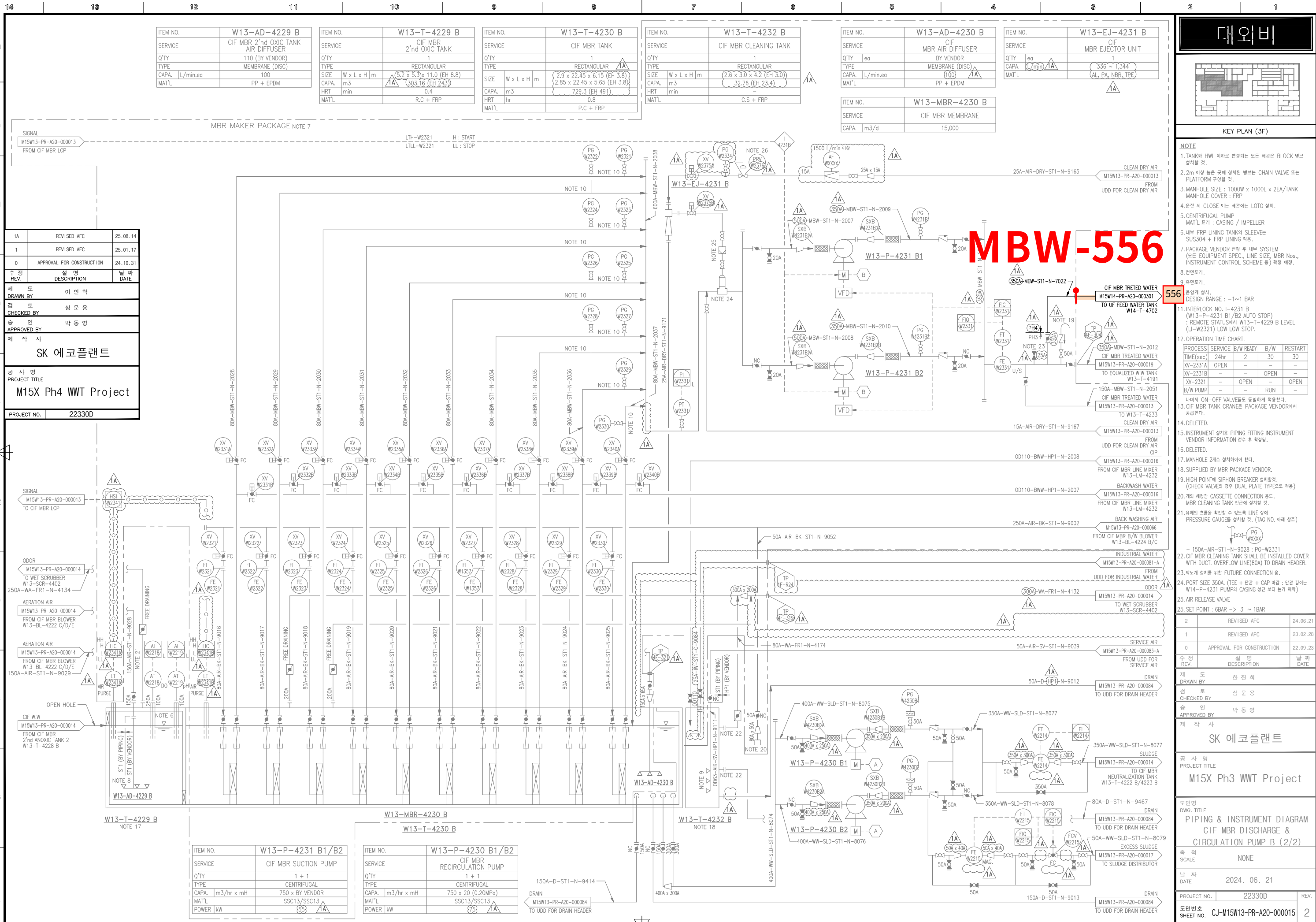
ITEM NO.		W14-MBR-4230 B
SERVICE		CIF MBR MEMBRANE
CAPA.	m3/d	15,000

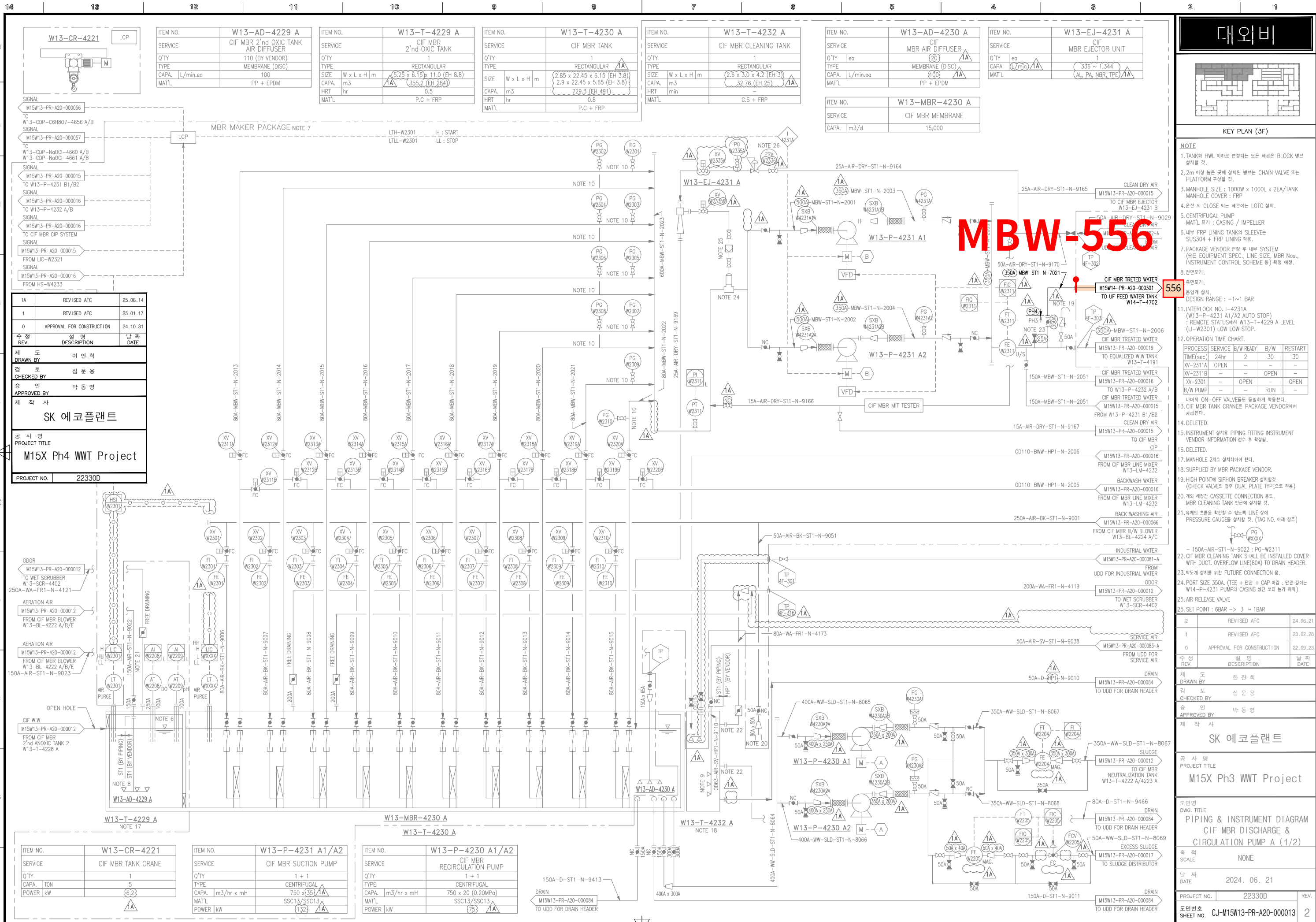
ITEM NO.	W14-EJ-4231B
SERVICE	CIF MBR EJECTOR UNIT
Q'TY ea	1
CAPA. NI/s.ea	BY VENDOR
MAT'L	BY VENDOR



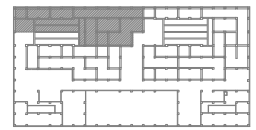
ITEM NO.	W14-P-4231 B1/B2		
SERVICE	CIF MBR SUCTION PUMP		
Q'TY	1 + 1		
TYPE	CENTRIFUGAL (VWF)		
CAPA.	m ³ /hr x mH	750 x 35	2A
MAT'L	SSC13/SSC13		
POWER	kW	132	2A

ITEM NO.		W14-P-4230 B1/B2	
SERVICE		CIF MBR RECIRCULATION PUMP	
Q'TY		1 + 1	
TYPE		CENTRIFUGAL	
CAPA.	m3/hr x mH	750 x 20 (0.20MPa)	
MAT'L		SSC13/SSC13	
POWER	kW	(90) / 2A	





대외비



KEY PLAN (3F)

NOTE

1. TANK의 HWL 이하로 연결되는 모든 배관은 BLOCK 밸브 설치할 것.
2. 2m 이상 높은 곳에 설치된 밸브는 CHAIN VALVE 또는 PLATFORM 구성할 것.
3. MANHOLE SIZE : 1000W x 1000L x 2EA/TANK MANHOLE COVER : FRP
4. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
5. CENTRIFUGAL PUMP MAT'L 표기 : CASING / IMPELLER
6. 내부 FRP LINING TANK에 SLEEVE SUS304 + FRP LINING 적용.
7. PACKAGE VENDOR 선정 후 내부 SYSTEM (모든 EQUIPMENT SPEC., LINE SIZE, MBR Nos., INSTRUMENT CONTROL SCHEME 등) 확정 예정.
8. 전연포기.
9. 유입계 설치. DESIGN RANGE : -1~1 BAR
11. INTERLOCK NO. I-4231A (W13-P-4231 A1/A2 AUTO STOP) : REMOTE STATUS에 W13-T-4229 A LEVEL (L-W2301) LOW LOW STOP.
12. OPERATION TIME CHART.

PROCESS	SERVICE	B/W READY	B/W	RESTART
TIME(sec)	24hr	2	30	30
XV-2311A	OPEN	-	-	-
XV-2311B	-	-	OPEN	-
XV-2301	-	OPEN	-	OPEN
B/W PUMP	-	-	RUN	-
- 나미지 ON-OFF VALVE도 동원하게 적용한다.
13. CIF MBR TANK CRANE에 PACKAGE VENDOR에서 공급한다.
14. DELETED.
15. INSTRUMENT 설치용 PIPING FITTING INSTRUMENT VENDOR INFORMATION 접수 후 확정함.
16. DELETED.
17. MANHOLE 2개소 설치하여야 한다.
18. SUPPLIED BY MBR PACKAGE VENDOR.
19. HIGH POINT에 SIPHON BREAKER 설치함. (CHECK VALVE의 경우 DUAL PLATE TYPE으로 적용)
20. 개회 세정간 CASSETTE CONNECTION 용도. MBR CLEANING TANK 인근에 설치할 것.
21. 유체의 흐름을 확인할 수 있도록 LINE 상에 PRESSURE GAUGE를 설치할 것. (TAG NO. 아래 참고)
- 150A-AIR-ST1-N-9022 : PG-W2311
22. CIF MBR CLEANING TANK SHALL BE INSTALLED COVER WITH DUCT. OVERFLOW LINE(80A) TO DRAIN HEADER.
23. 막도개 설치를 위한 FUTURE CONNECTION 용.
24. PORT SIZE 350A. (TEE + 핀관 + CAP 매립 : 단면 길이는 W14-P-4231 PUMP의 CASING 상단 보다 높게 제작)
25. AIR RELEASE VALVE
25. SET POINT : 6BAR -> 3 ~ 1BAR

2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 도	한 진 희	
검 토	심 문 용	
승 인	박 동 영	
제 작 사	SK 에코플랜트	
공 사 명	PROJECT TITLE	
M15X Ph3 WWT Project		
도면명	DWG. TITLE	
PIPING & INSTRUMENT DIAGRAM		
CIF MBR DISCHARGE & CIRCULATION PUMP A (1/2)		
축척	SCALE	
NONE		
날짜	DATE	
2024. 06. 21		
PROJECT NO.	22330D	REV.
도면번호	SHEET NO. CJ-M15W13-PR-A20-000013	2



1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB2/VB2L0	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDL0	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT



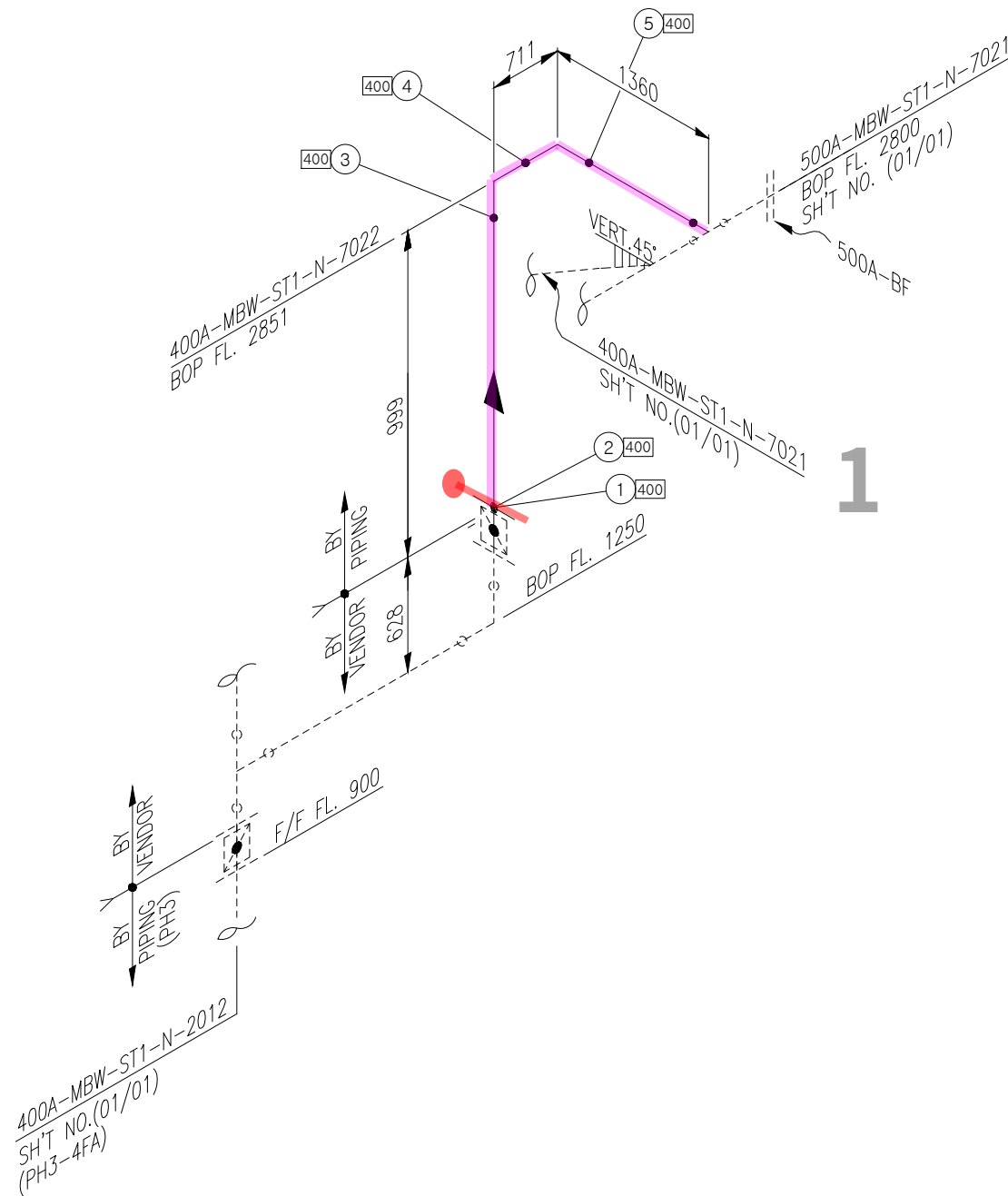
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	BY VENDOR	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	BY VENDOR	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040005-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D	
						REF. P & ID		REV.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000301								400A-MBW-ST1-N-7021	01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	400A	1.5M	PIPE, STS 304TP, ERW, BE – SCH 10S
E90	400A	2EA	ELBOW 90 DEG SR, STS304, ERW, BE – SCH 10S
F	400A	1EA	FLG SO, KS 10K, RF, STS304
G	400A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	400A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



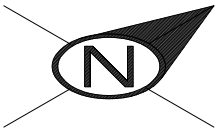
<NOTE>

1. (*) FIELD ADJUST

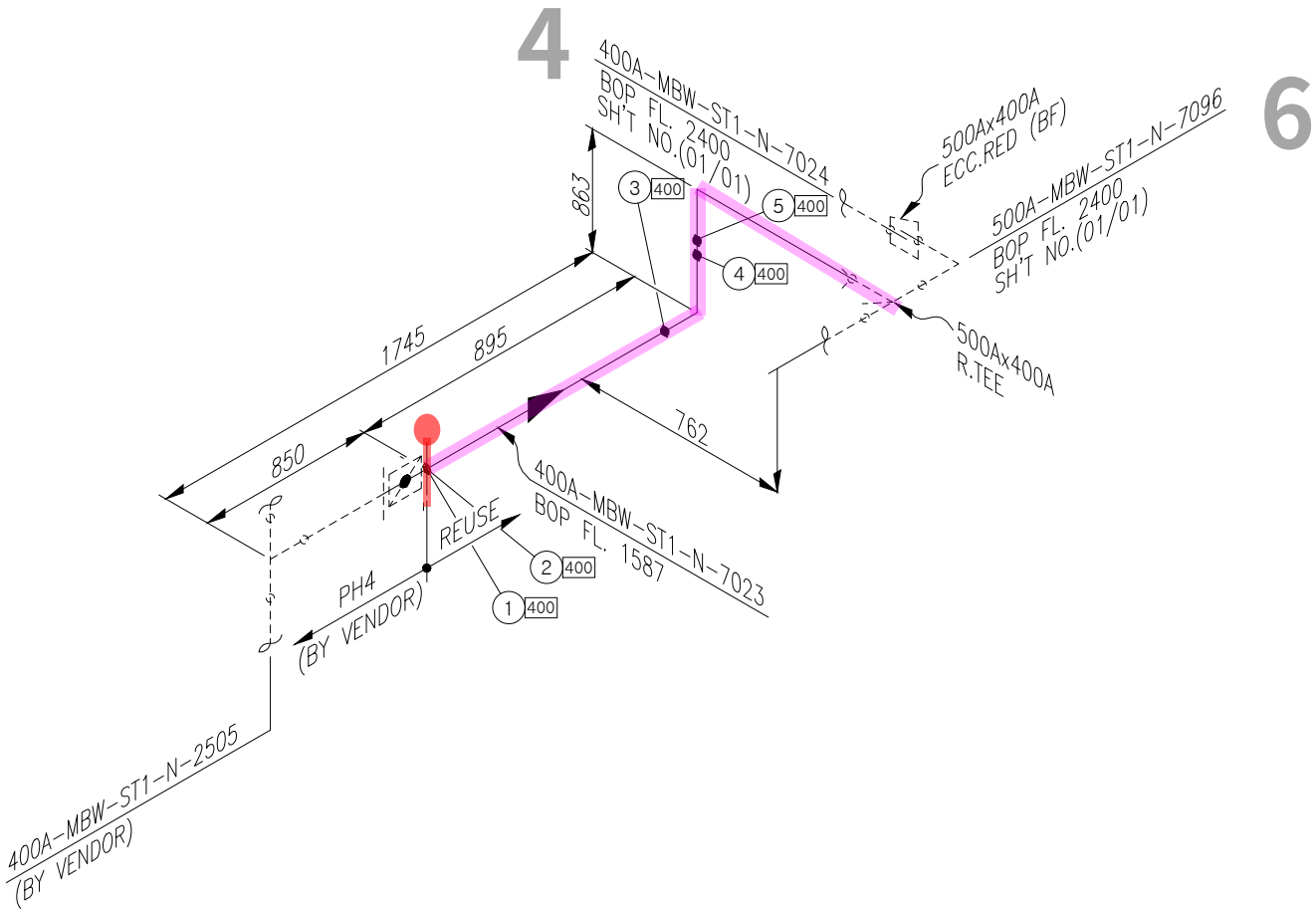
* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	BY VENDOR	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	BY VENDOR	AMB	REF. PLAN DWG.	CJ-M15W14-PI-A30-040005-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	MBW	JOB NO.	
						REF. P & ID	CJ-M15W14-PR-A20-000301							REV. NO.	DATE	DESCRIPTION	BY

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	400A	0.9M	PIPE, STS 304TP, ERW, BE - SCH 10S
E90	400A	2EA	ELBOW 90 DEG SR, STS304, ERW, BE - SCH 10S
F	400A	1EA	FLG SO, KS 10K, RF, STS304
G	400A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	400A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



<NOTE>

1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT



3

CONTRACTOR



PROJECT NAME

M15X Ph4 Project

COMMODITY

MBW

JOB NO.

22330D

LINE NO.

400A-MBW-ST1-N-7023

SH'T NO.

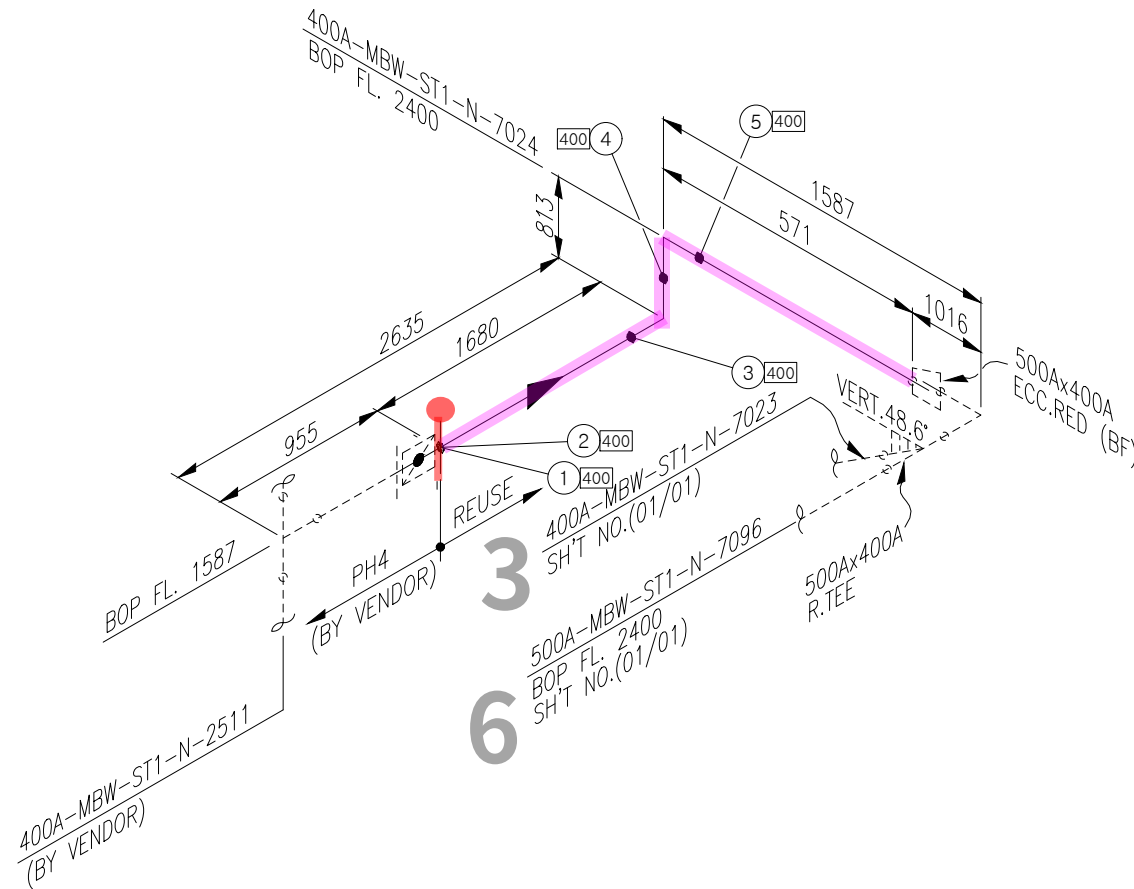
01 / 01

REV.

0



ITEM	SIZE	Q'TY	DESCRIPTION
ST1			
P	400A	1.5M	PIPE, STS 304TP, ERW, BE – SCH 10S
E90	400A	2EA	ELBOW 90 DEG SR, STS304, ERW, BE – SCH 10S
F	400A	1EA	FLG SO, KS 10K, RF, STS304
G	400A	1EA	GASKET, KS 10K, RF, PTFE+EPDM, 3.0MM TH'K
BL	400A	1SET	MACHINE BOLT & NUTS :STS 304 / W WASHERS(SPECIAL)



<NOTE>

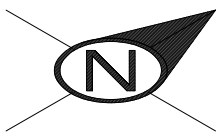
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

CLIENT		4	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
MBW		22330D	
LINE NO.	400A-MBW-ST1-N-7024	SH'T NO.	REV.
		01 / 01	0

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	BY VENDOR	60	UNIT	4FA						
			OPERATING	BY VENDOR	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-040006-A							
						REF. P & ID							
						CJ-M15W14-PR-A20-000301							
								0	25.01.24	APPROVED FOR CONSTRUCTION	황래종	안남후	박동영
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음

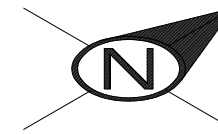


1. (*) FIELD ADJUST

CONTRACTOR 		PROJECT NAME	
COMMODITY MBW		M15X Ph4 Project	
LINE NO. 500A-MBW-ST1-N-7021		JOB NO. 22330D	
		SH'T NO. 01 / 01	REV. 0

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	BY VENDOR	60	UNIT 4FA						
			OPERATING	BY VENDOR	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040005-A						
						REF. P & ID CJ-M15W14-PR-A20-000301		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D

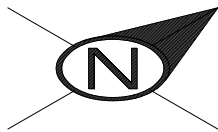
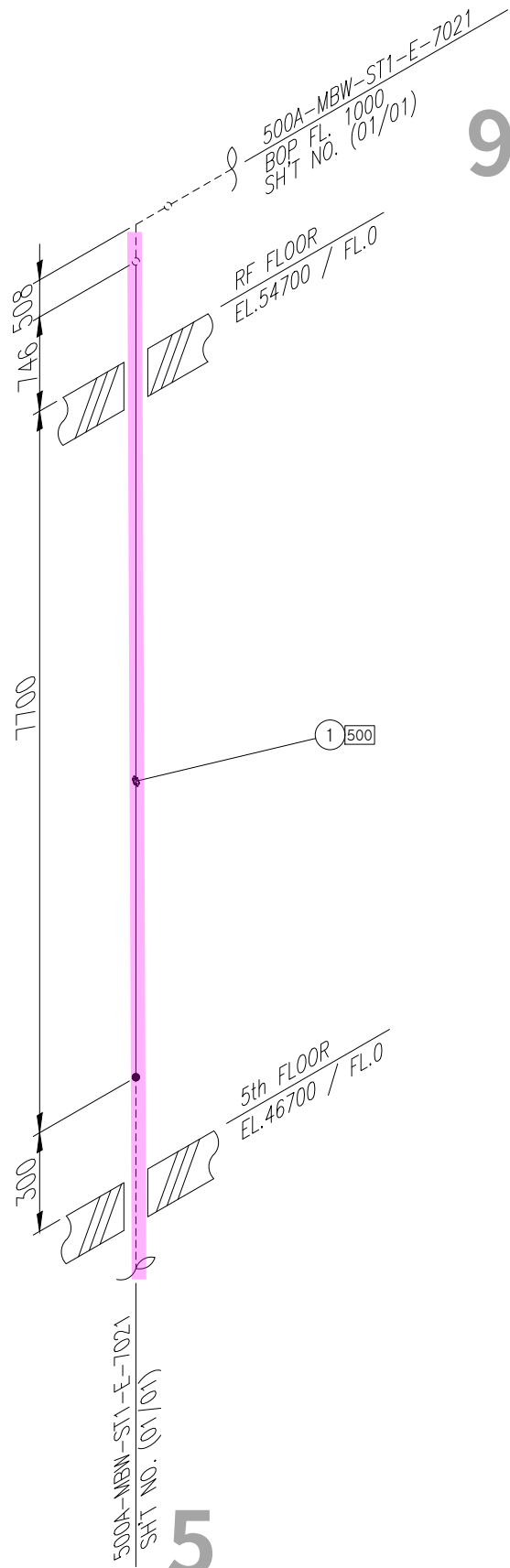
당고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유보자의 불법 사용으로 피해가 발생할 경우 유보자에게 민/형사상 책임을 요구할 수 있음



1. (*) FIELD ADJUST

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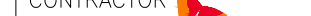
SH'T NO. 01 / 01	REV. 0
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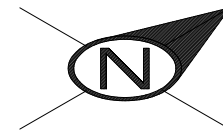
<NOTE>

1. (*) FIELD ADJUST

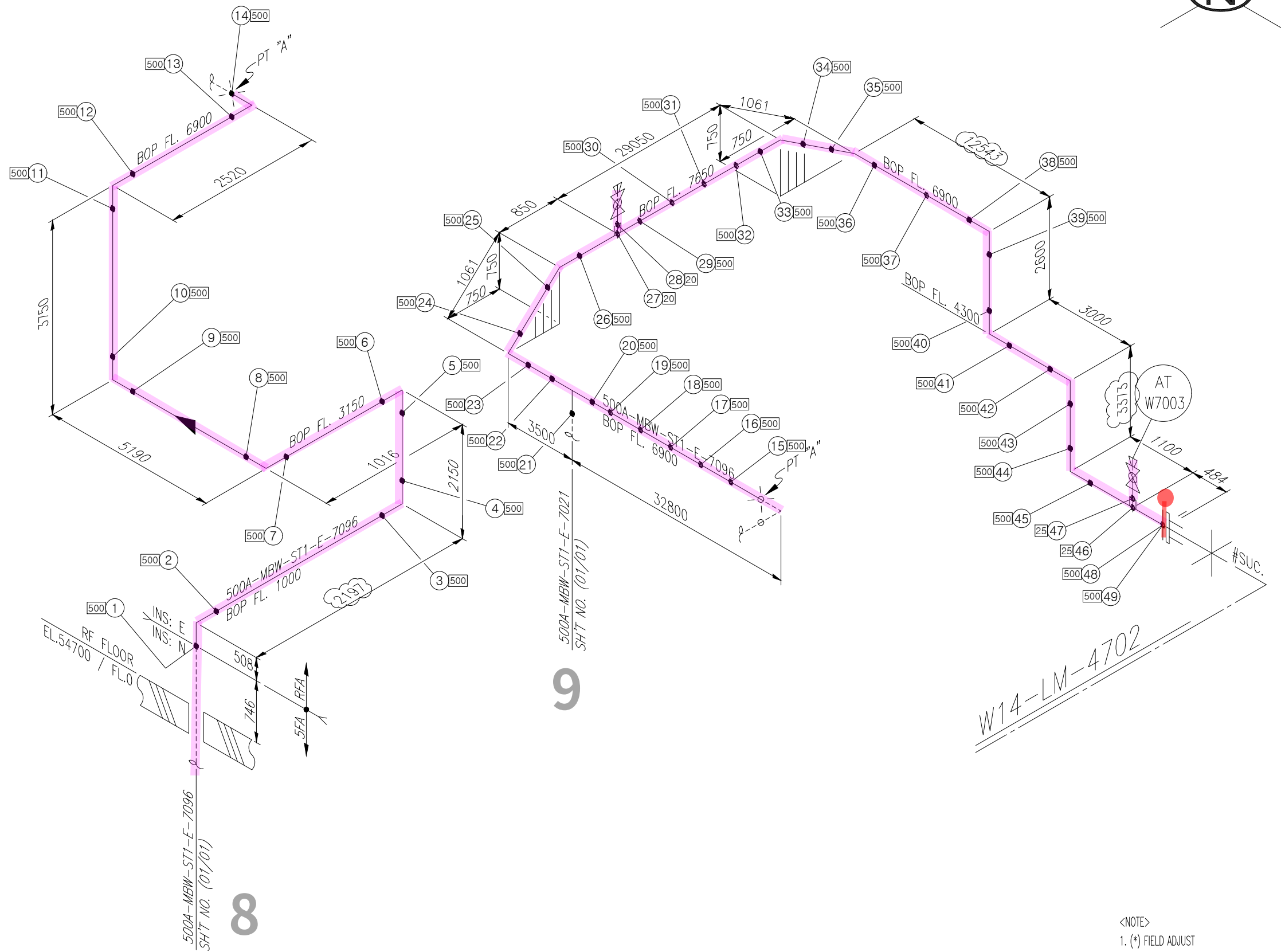
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14									CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	0.8	60	UNIT	5FA										M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-05006-A										COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000301		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영				LINE NO.	500A-MBW-ST1-N-7021	SH'T NO. 01 / 01	REV. 0





<NOTE>
1. (*) FIELD ADJUST



<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	E	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1	60	UNIT	RFA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-R10007-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY		JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000301			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	MBW		22330D	
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	500A-MBW-ST1-E-7096	SH'T NO. 01 / 01	REV. 1

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음.

SK ecoplant

CLIENT

SK hynix

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CONTRACTOR

SK ecoplant

COMMODITY
MBW

LINE NO.
500A-MBW-ST1-E-7096

PROJECT NAME

M15X Ph4 Project

JOB NO.

22330D

SH'T NO.

01 / 01

REV.

1